

Ethanol Blending in Gasoline – Slovakia

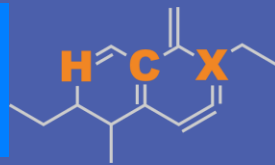
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**U.S. GRAINS &
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Ethanol Blending in Europe

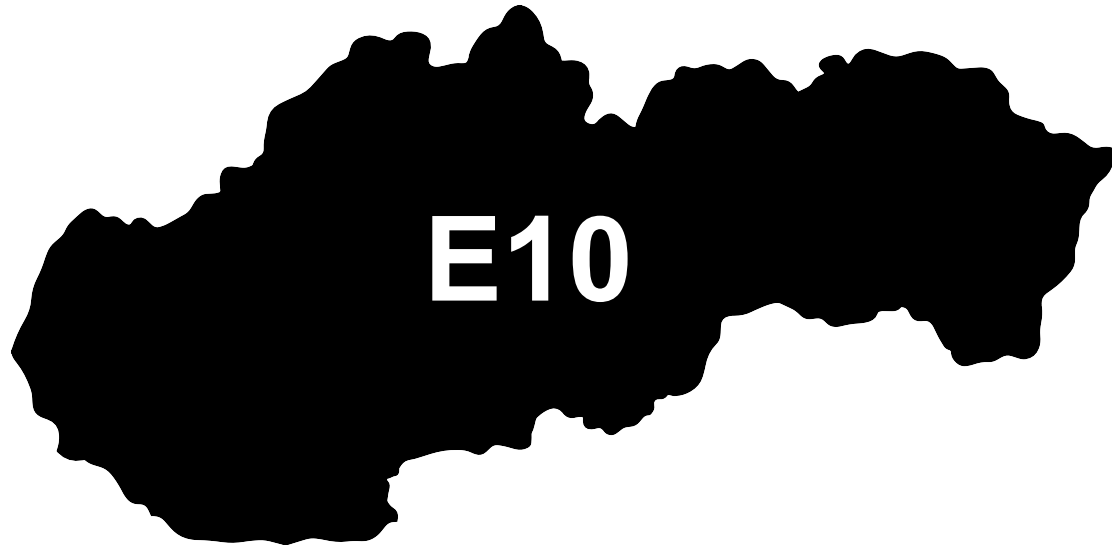
There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

Ethanol Gasoline Blending in Slovakia

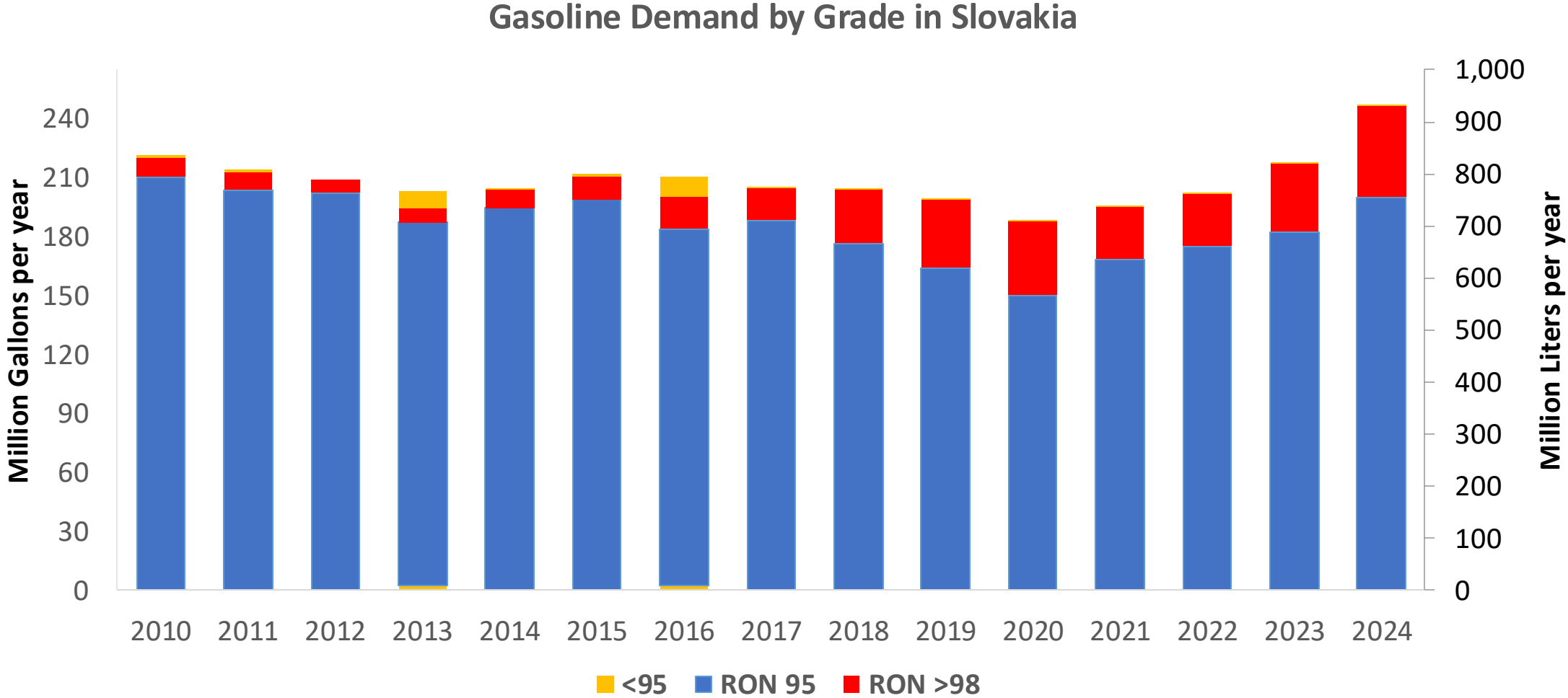


In 2024, gasoline consumption in Slovakia was 931 million liters (246 million gallons) and growth rate was 4.4%. Gasoline production and net imports were and 931 million liters (and 246 million gallons) correspondingly. Slovakia complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.6 RON.

In 2024, ethanol consumption in Slovakia was 80 million liters (21 million gallons) representing 8.6% of gasoline demand. Ethanol production and Net Imports were 80 and million liters (21 and million gallons) correspondingly. Ethanol source is Cereals 100%.

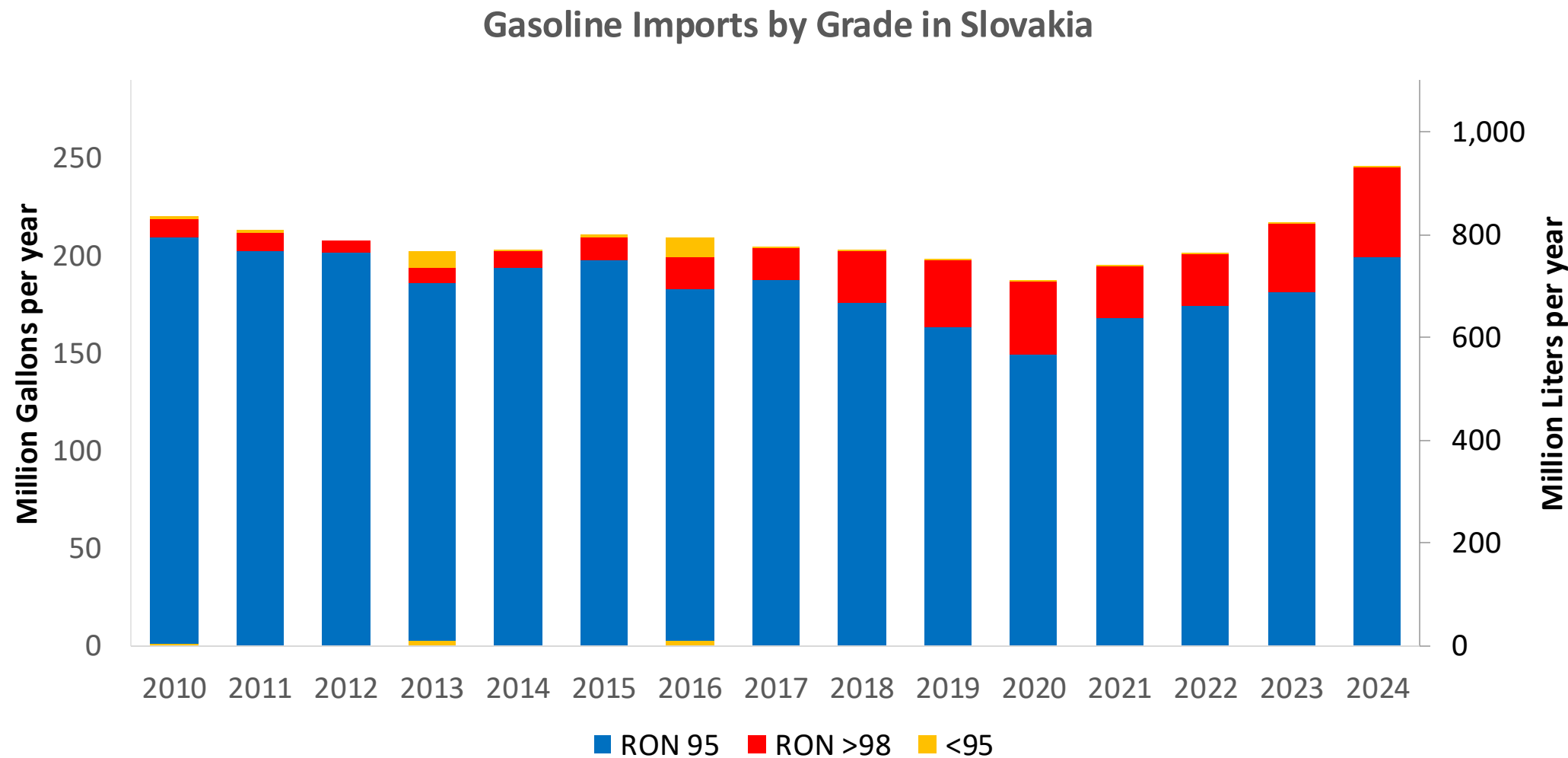
Source: Eurostat, European Commission

Gasoline Demand in Slovakia



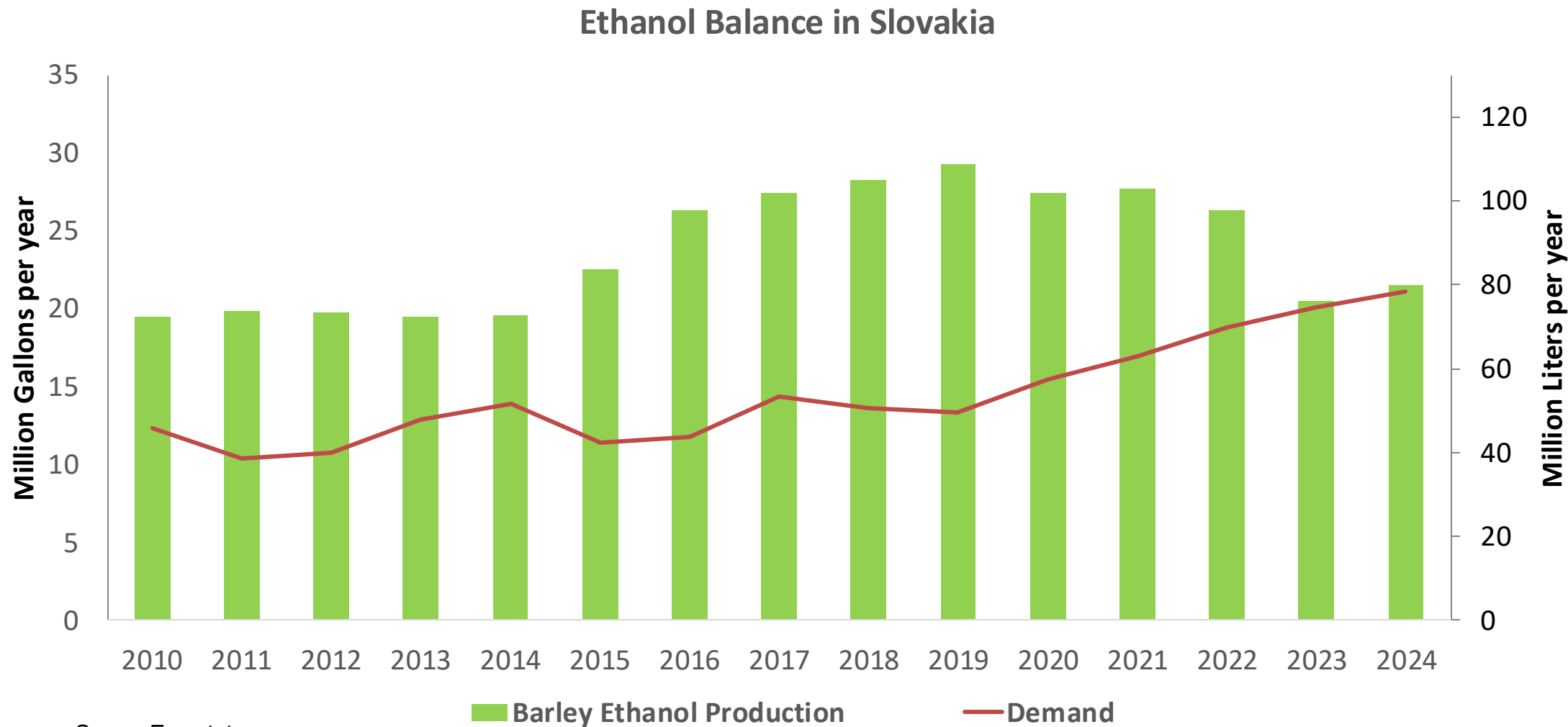
Source: Eurostat

Gasoline Imports in Slovakia



Source: Eurostat

Ethanol Balance in Slovakia



Source: Eurostat

Gasoline Quality in Slovakia

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Countries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	< 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	9.2							
Advanced Biofuels (%cal)	1.1							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commission

The FARO90 logo is shown in a blue box. To its right is a chemical structure of a polyene chain with a terminal group labeled 'X'.

0% 10% 20% 30% 40% 50% 60% 70% 80% 90% 100%



Source: Faro90

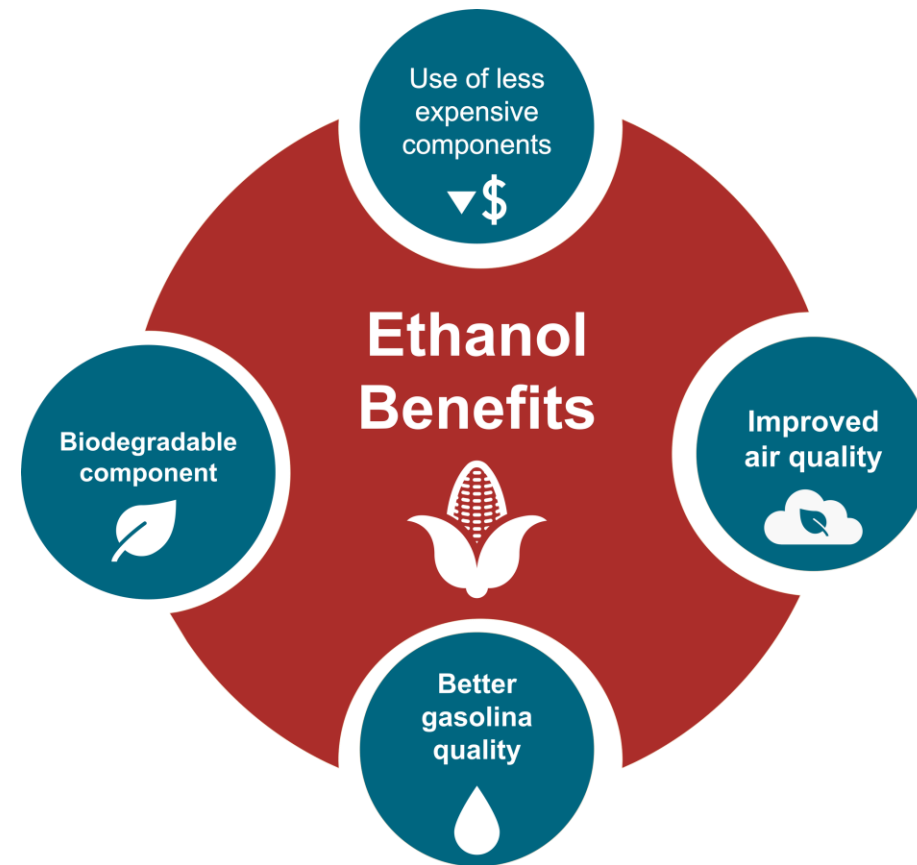
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

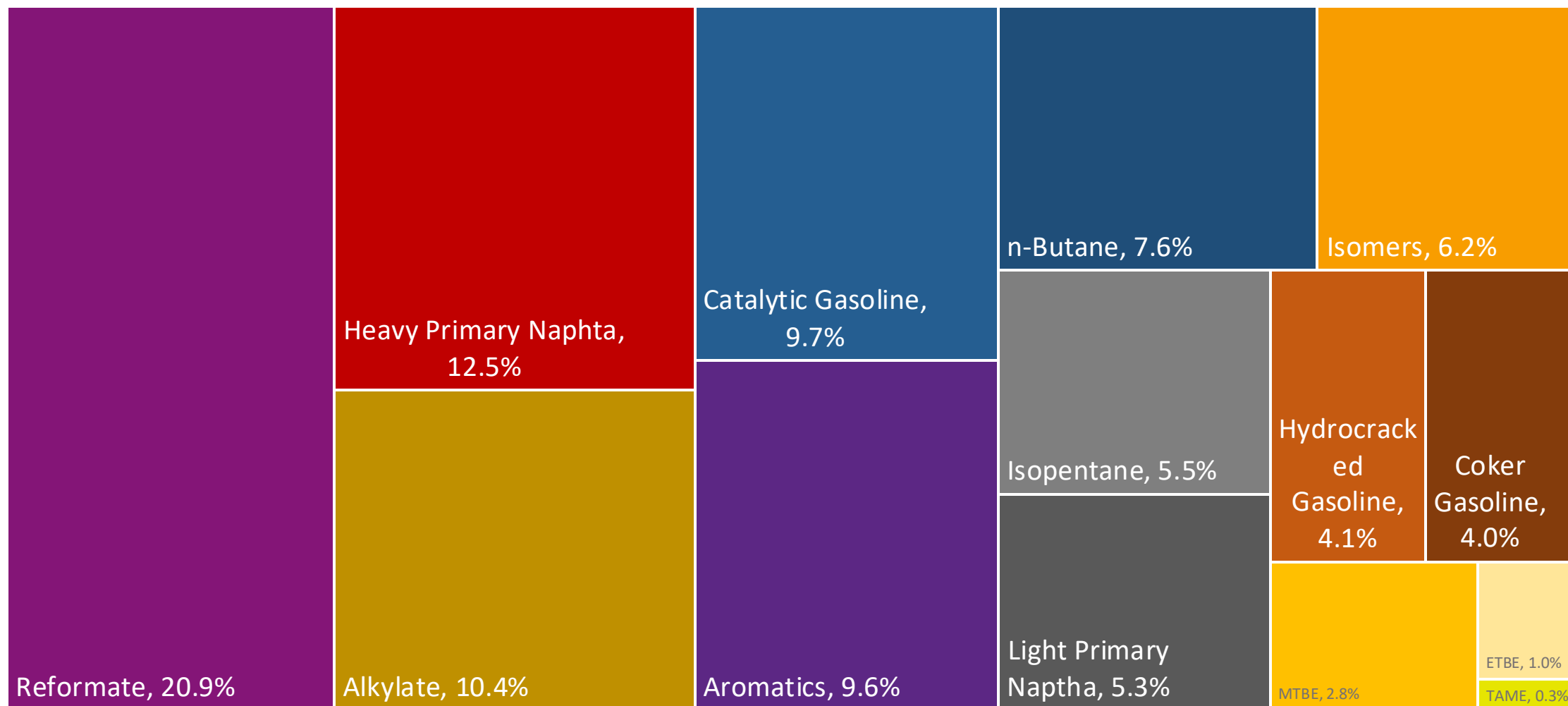
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

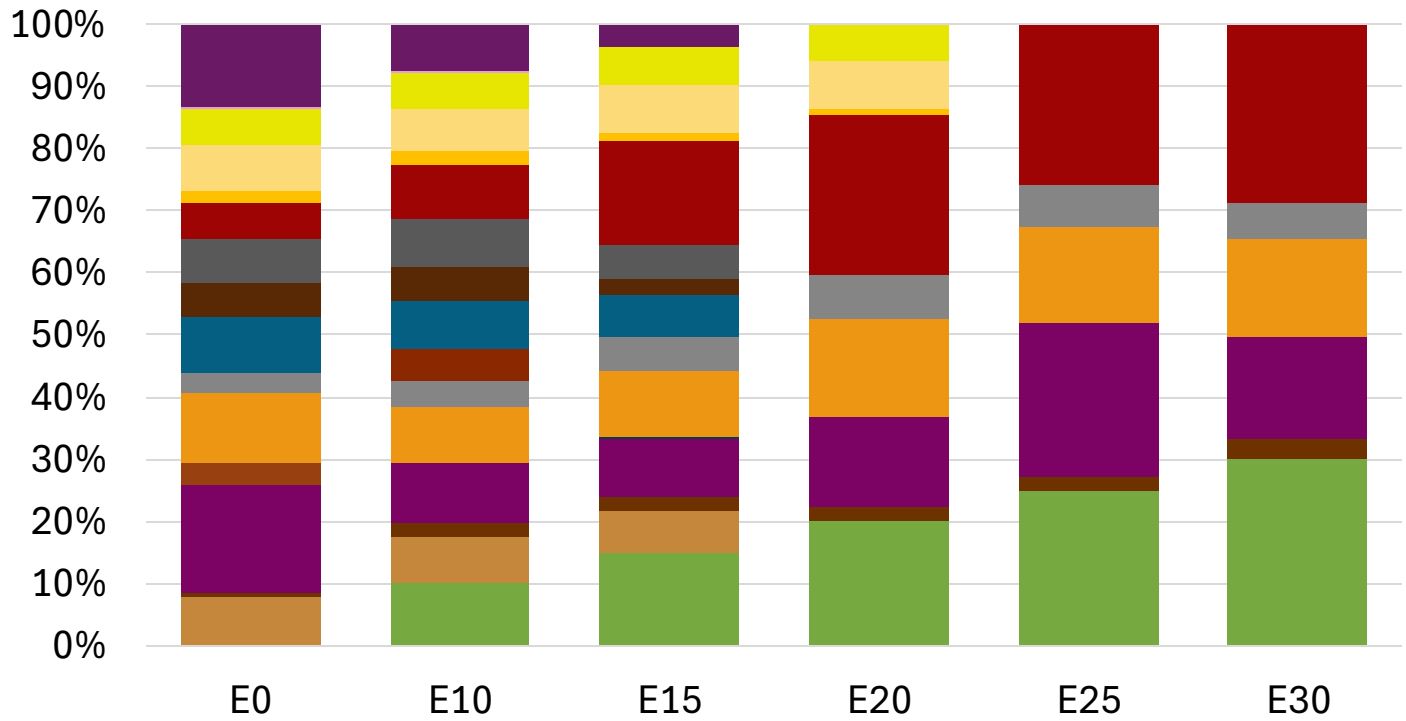
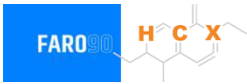
The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Slovakia Available Blending Components



Slovakia – Gasoline 95 – Constant Octane

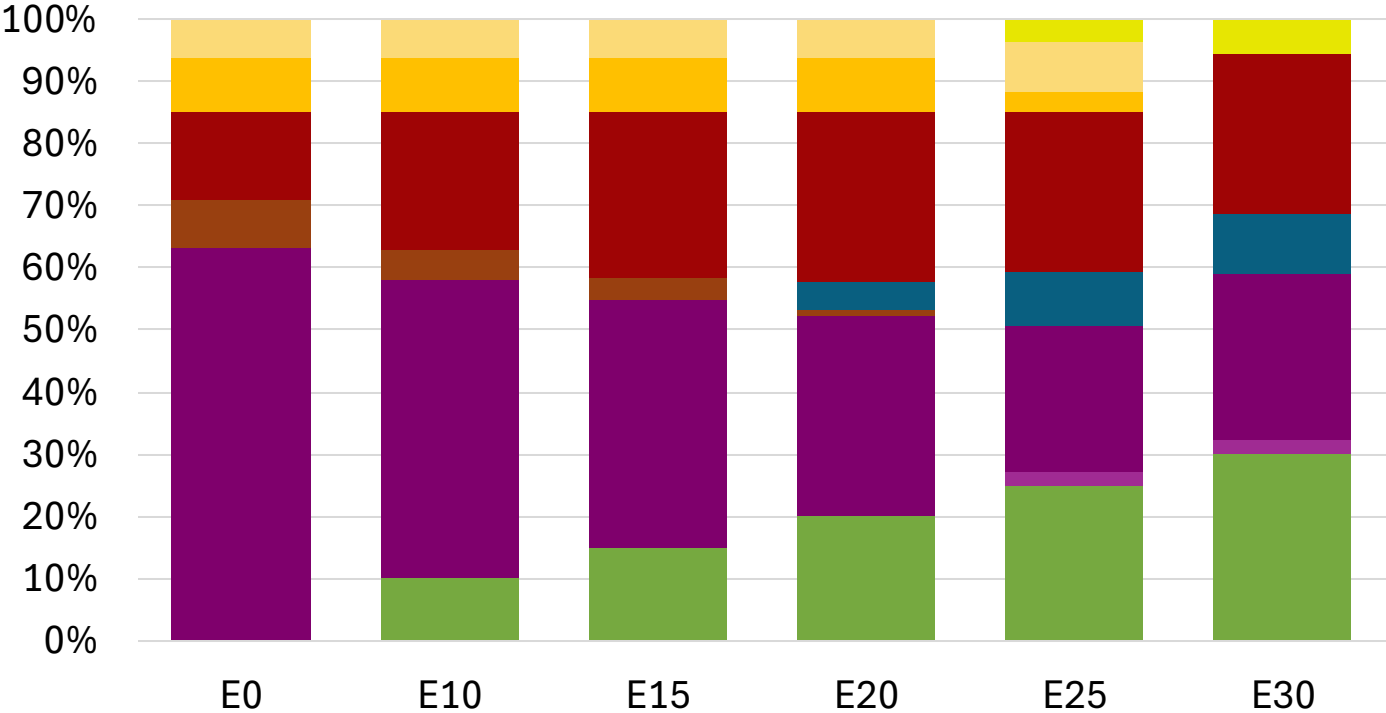
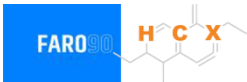


Average CIF 2024 Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US\$/gal)	\$ 2.304	\$ 2.166	\$ 2.066	\$ 1.951	\$ 1.905	\$ 1.852

Source: Faro90

Slovakia - Gasoline 98 - Constant Octane

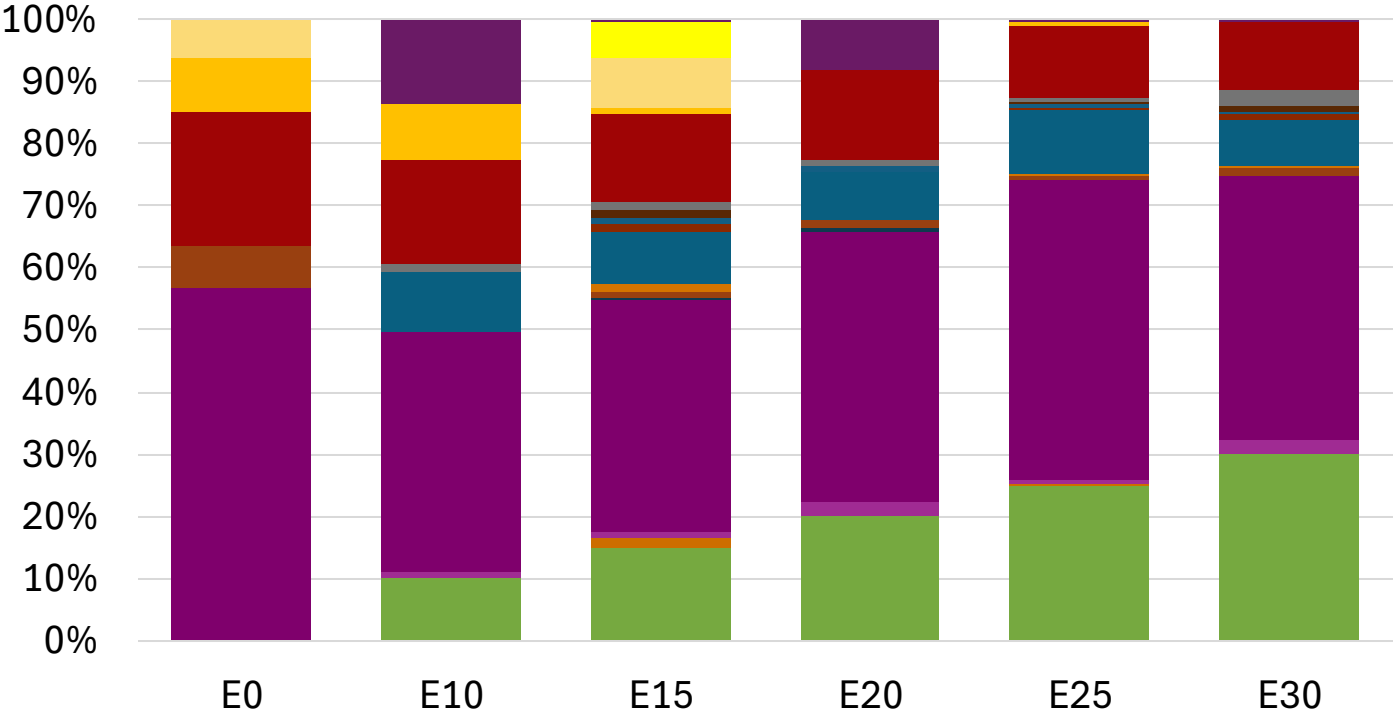


Average CIF 2024 Prices

Octane (RON)	98.0					
Price (US\$/gal)	\$ 2.324	\$ 2.207	\$ 2.138	\$ 2.066	\$ 2.001	\$ 1.951

Source: Faro90

Slovakia – Gasoline 95 – Increased Octane

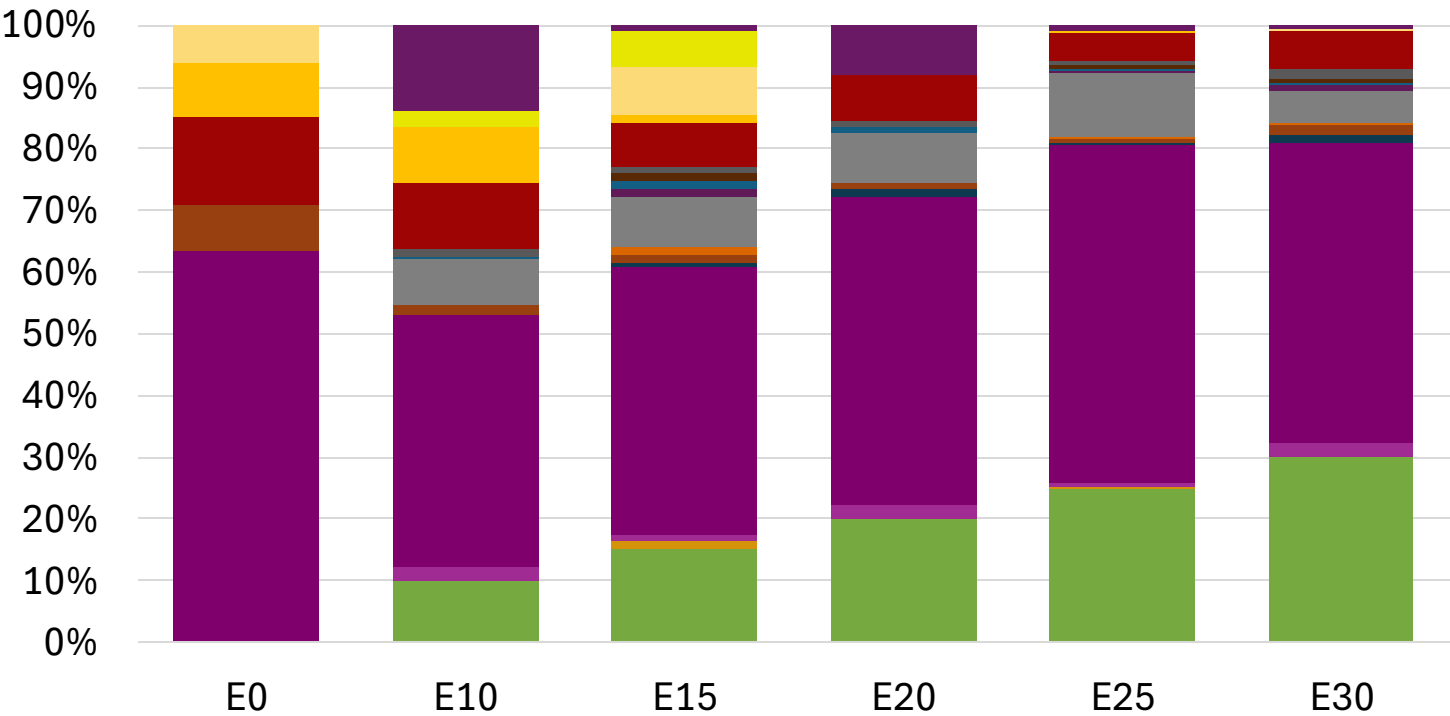


Average CIF 2024 Prices

Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US\$/gal)	\$ 2.192	\$ 2.192	\$ 2.192	\$ 2.192	\$ 2.192	\$ 2.192

Source: Faro90

Slovakia – Gasoline 98 – Increased Octane



Average CIF 2024
Prices

Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US\$/gal)	\$ 2.324	\$ 2.324	\$ 2.324	\$ 2.324	\$ 2.324	\$ 2.324

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

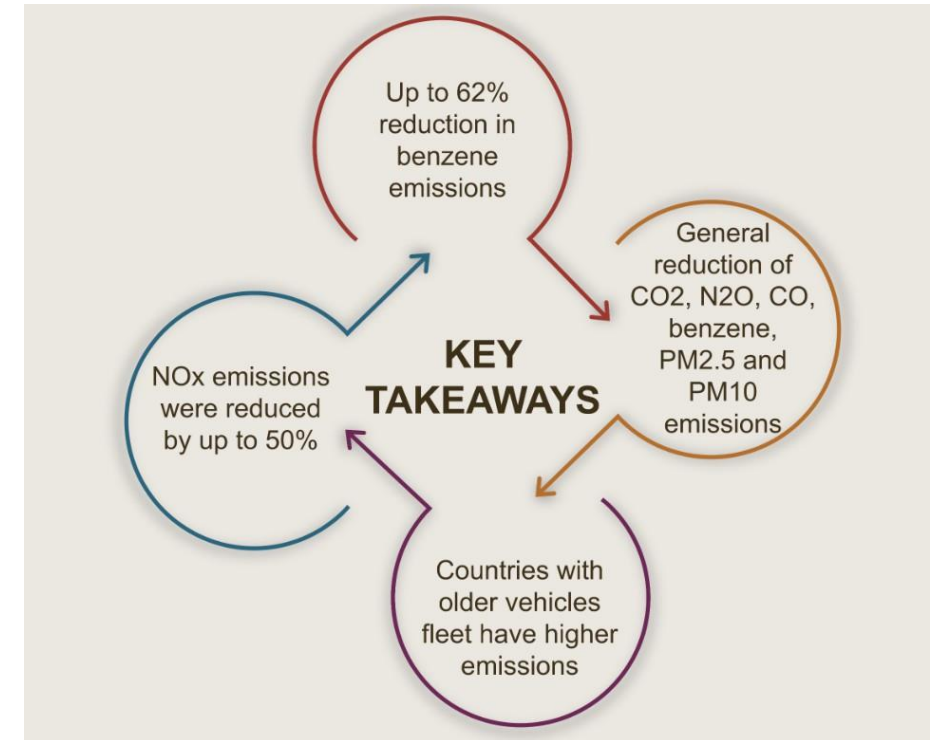
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

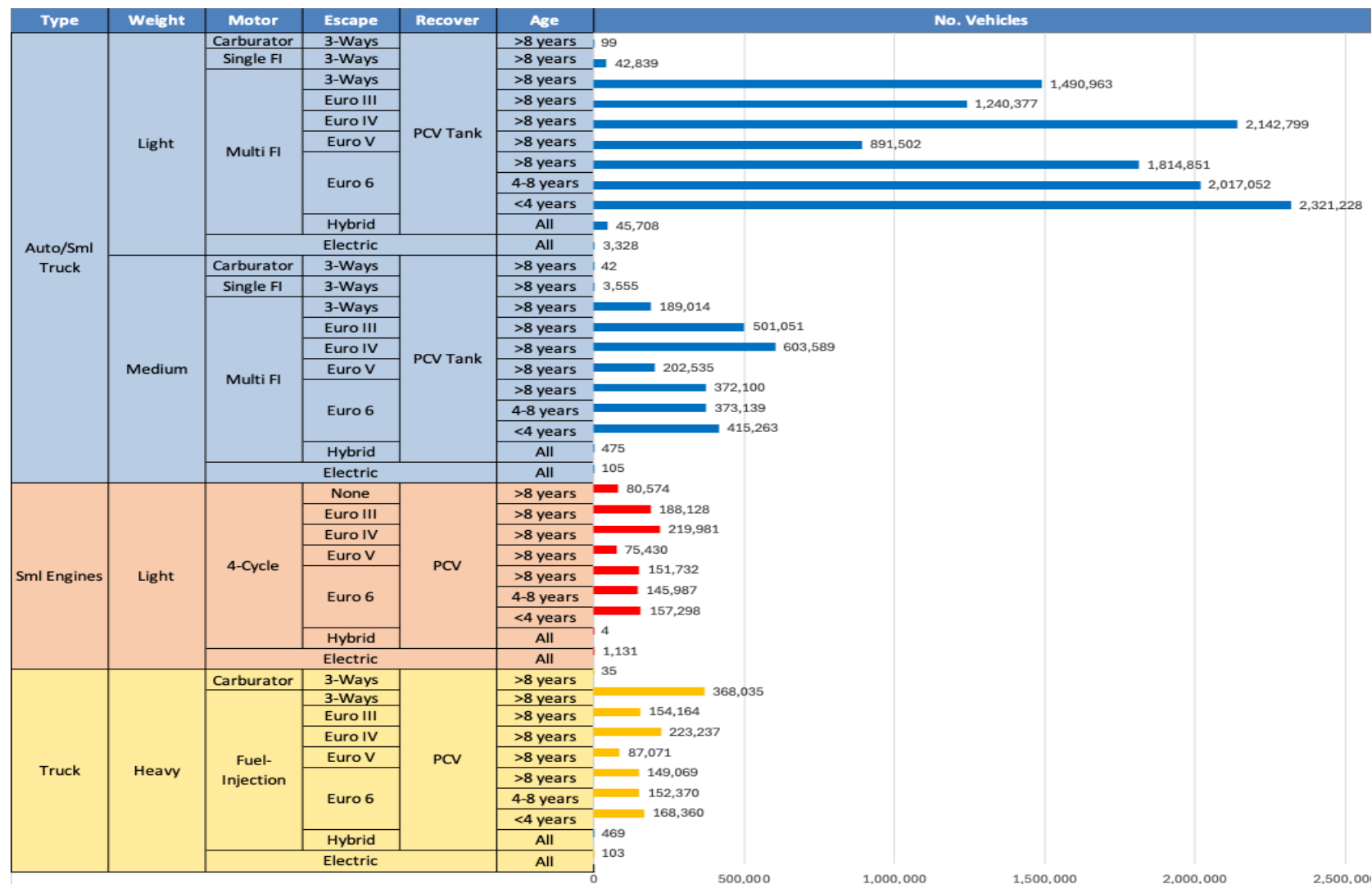
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

**Source: Bureau of transportation statistics.*

Main Results



Gasoline Vehicle Fleet - Slovakia

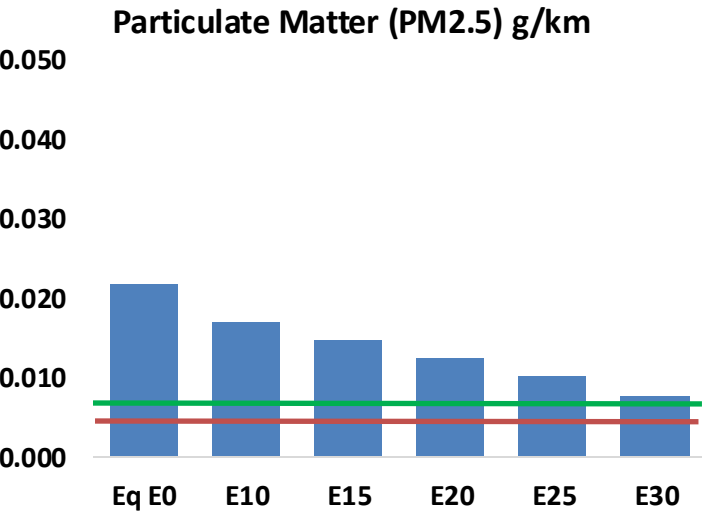
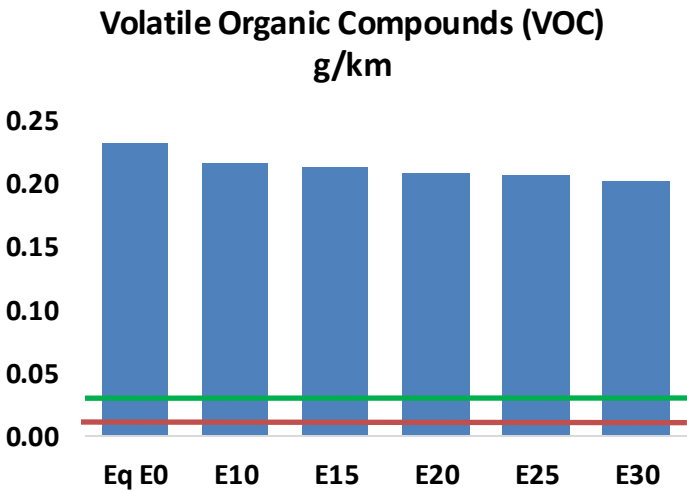
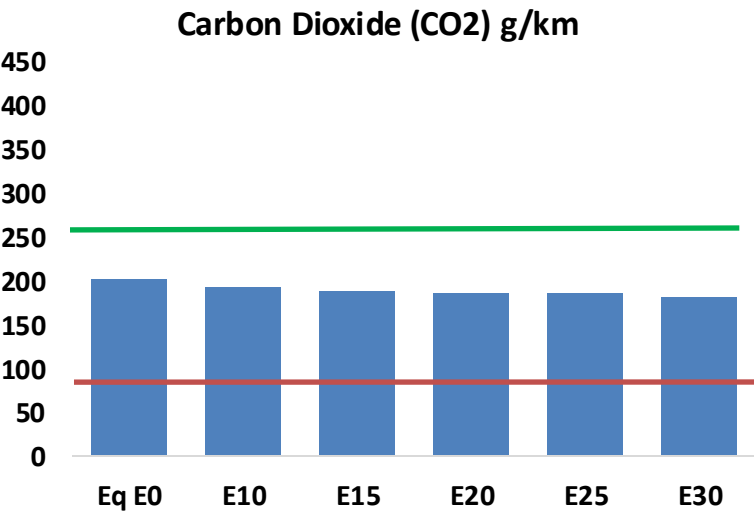
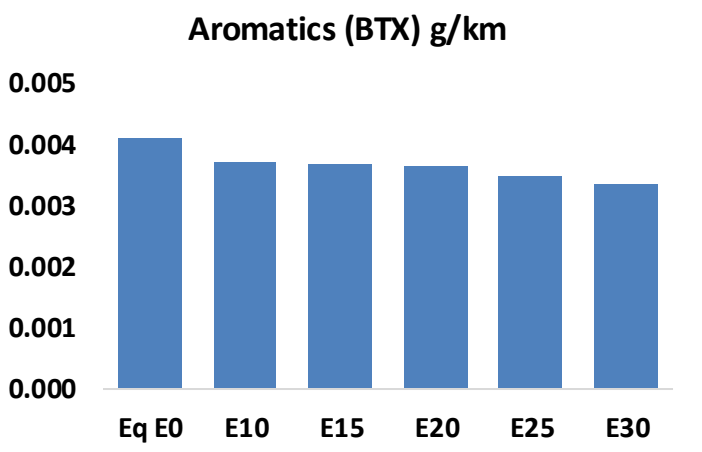
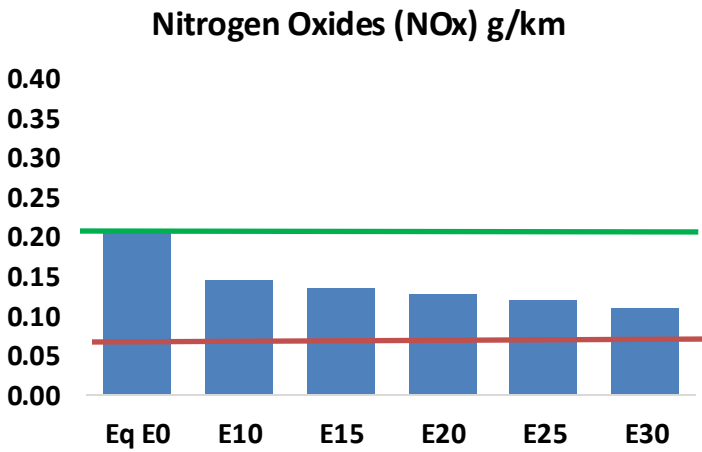
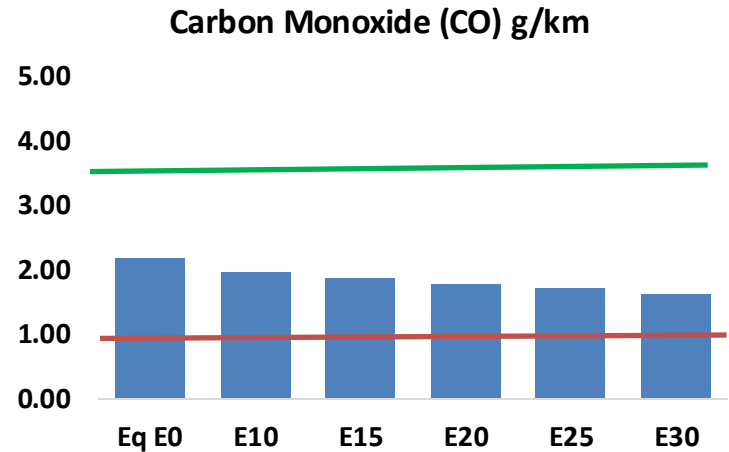
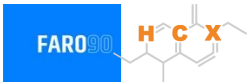


Vehicle Fleet: 5,792,410
Gasoline Vehicle Fleet: 1,843,374

Source: Eurostat, ACEA

Slovakia – Vehicular Emissions

TIER III BIN 125 United States
Euro 6 Standard



Slovakia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	99,865	94,329	88,793	84,675	80,722	77,741	74,129	-11%	-19%
CO2	9,214,229	8,977,292	8,753,517	8,577,526	8,490,438	8,440,370	8,284,790	-5%	-8%
VOC	10,576	10,223	9,870	9,671	9,511	9,401	9,210	-7%	-10%
NOx	9,439	7,079	6,608	6,228	5,873	5,482	5,069	-30%	-38%
SOx	104	96	88	81	75	68	61	-15%	-28%
NH3	3,205	3,174	3,142	3,157	3,192	3,217	3,238	-2%	0%
N2O	103	102	102	103	105	106	107	-1%	2%
CH4	2,586	2,586	2,586	2,625	2,681	2,731	2,777	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	109	106	103	102	102	101	100	-5%	-7%
Acetaldehyde	233	267	392	597	813	929	1,098	68%	249%
Formaldehyde	744	724	844	991	1,038	1,148	1,250	13%	39%
Benzene	187	179	170	167	166	159	153	-9%	-11%
PM2.5	590	524	458	398	338	274	208	-22%	-43%
PM10	403	358	313	272	231	187	142	-22%	-43%

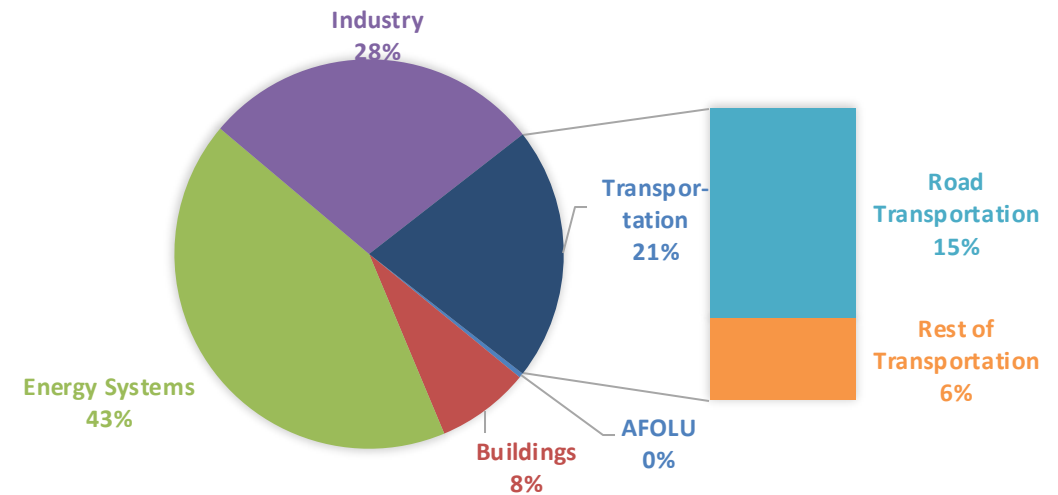
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

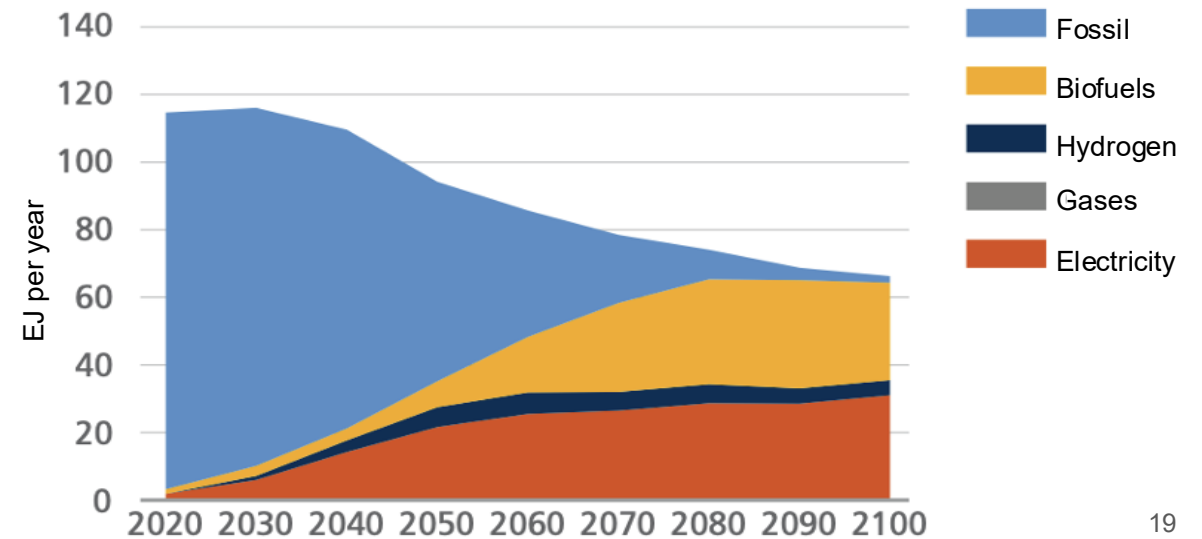
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

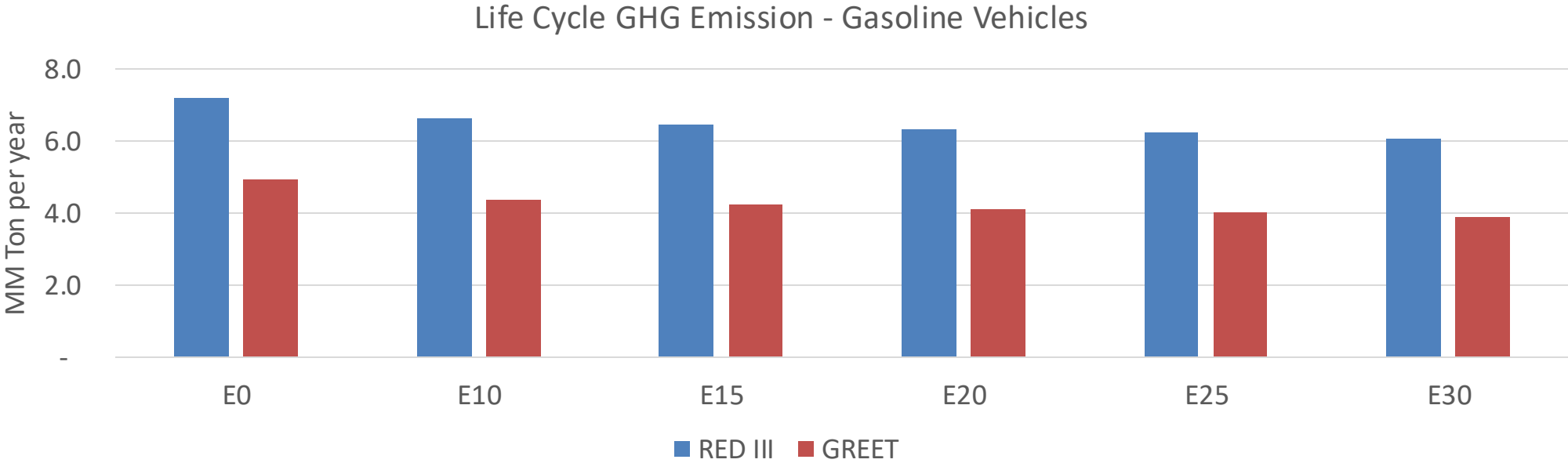
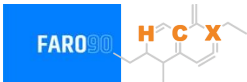
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Slovakia – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	36.8
Target @ 2035 (MMT/year)	20.7
Delta	16.1

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	11.2%	14.2%	16.5%	18.6%	21.6%
RED II/III	0.0%	8.1%	10.3%	12.0%	13.5%	15.7%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	3.4%	4.4%	5.0%	5.7%	6.6%
RED II/III	0.0%	3.6%	4.6%	5.4%	6.1%	7.0%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90